Problem A. Solve Me First

OS Linux

Read in two integers from stdin, add them together, and print the answer to stdout.

Input Format

There are 2 lines of input. Each line contains one integer.

Output Format

Print the sum of the integers on one line.

Note: Everything your program prints is considered "output", so if you output some debugging statements like "Please enter an integer:", this will be considered as part of your answer, and because it does not satisfy the output format, it will be marked wrong, even if your answer is otherwise correct!

Sample Input

```
3
4
```

Sample Output

```
7
```

Problem B. Simple Array Sum

OS Linux

Given an array of integers, find the sum of its elements.

For example, if the array is $[1, 2, 3]$, then the sum is $1 + 2 + 3 = 6$, so output `6`.

Input

The first line contains the integer n denoting the length of the array.

The second line contains n space-separated integers representing the array's elements.

Output

Print a single integer: the sum of the elements of the array.

Constraints

$0 \leq n, \text{each element} \leq 1000$

Sample Input

```
6
1 2 3 4 10 11
```

Sample Output

```
31
```

Problem C. A Very Big Sum

OS Linux

Calculate and print the sum of the elements in an array.

Input Format

The first line of the input consists of an integer n.
The next line contains n space-separated integers contained in the array.

Output Format

Print the integer sum of the elements in the array.

Constraints

$1 \leq n \leq 10$

$0 \leq ar[i] \leq 10^{10}$

Sample Input

5
1000000001 1000000002 1000000003 1000000004 1000000005

Output

5000000015

Note:

The range of the 32-bit integer is **-2147483648 to 2147483647**.

When we add several integer values, the resulting sum might exceed the above range. You might need to use long long int (%lld) C/C++ to store such sums.

Problem D. Missing Number

Time Limit 1000 ms
Mem Limit 524288 kB

Given the sum of four numbers and three of those numbers, find the missing number.

All four numbers are positive integers.

Input

The input will contain one integer (the sum of the four numbers) in the first line and three integers (of those four numbers) in the second line.

All numbers in this problem fit in 32-bit signed integers.

Output

Print the missing number.

Example

Input	Output
10 1 2 3	4

Problem E. Compare the Triplets

OS Linux

Alice and Bob each created one problem for HackerRank. A reviewer rates the two challenges, awarding points on a scale from *1* to *100* for three categories: *problem clarity*, *originality*, and *difficulty*.

The rating for Alice's challenge is the triplet *a = (a[0], a[1], a[2])*, and the rating for Bob's challenge is the triplet *b = (b[0], b[1], b[2])*.

The task is to calculate their comparison points by comparing each category:

- If *a[i] > b[i]*, then Alice is awarded *1* point.
- If *a[i] < b[i]*, then Bob is awarded *1* point.
- If *a[i] = b[i]*, then neither person receives a point.

Example

a = [1, 2, 3]
b = [3, 2, 1]

- For elements *0*, Bob is awarded a point because *a[0] < b[0]*.
- For the equal elements *a[1]* and *b[1]*, no points are earned.
- Finally, for elements *2*, *a[2] > b[2]* so Alice receives a point.

The return array is *[1, 1]* with Alice's score first and Bob's second.

Function Description

Complete the function *compareTriplets* with the following parameter(s):

- *int a[3]*: Alice's challenge rating
- *int b[3]*: Bob's challenge rating

Returns

- *int[2]*: the first element is Alice's score and the second is Bob's score

Input Format

The first line contains *3* space-separated integers, *a[0]*, *a[1]*, and *a[2]*, the respective values in triplet *a*.
The second line contains *3* space-separated integers, *b[0]*, *b[1]*, and *b[2]*, the respective values in triplet *b*.

Constraints

- $1 \le a[i] \le 100$
- $1 \le b[i] \le 100$

Input	Output
5 6 7 3 6 10	1 1

Explanation 0

In this example:

- $a = (a[0], a[1], a[2]) = (5, 6, 7)$
- $b = (b[0], b[1], b[2]) = (3, 6, 10)$

Now, let's compare each individual score:

- $a[0] > b[0]$, so Alice receives 1 point.
- $a[1] = b[1]$, so nobody receives a point.
- $a[2] < b[2]$, so Bob receives 1 point.

Alice's comparison score is 1, and Bob's comparison score is 1. Thus, we return the array $[1, 1]$.

Input	Output
17 28 30 99 16 8	2 1

Explanation 1

Comparing the 0th elements, $17 < 99$ so Bob receives a point.
Comparing the 1st and 2nd elements, $28 > 16$ and $30 > 8$ so Alice receives two points.
The return array is $[2, 1]$.

Problem F. Diagonal Difference

OS Linux

Given a square matrix, calculate the absolute difference between the sums of its diagonals.

For example, the square matrix arr is shown below:

```
1 2 3
4 5 6
9 8 9
```

- The left-to-right diagonal = $1 + 5 + 9 = 15$.
- The right-to-left diagonal = $3 + 5 + 9 = 17$.

Their absolute difference is $|15 - 17| = 2$.

Function description

Complete the $diagonalDifference$ function with the following parameter:

- $int\ arr[n][m]$: a 2-D array of integers

Return

- int: the absolute difference in sums along the diagonals

Input Format

The first line contains a single integer, n, the number of rows and columns in the square matrix arr.
Each of the next n lines describes a row, $arr[i]$, and consists of n space-separated integers $arr[i][j]$.

Constraints

- $-100 \leq arr[i][j] \leq 100$

Input	Output
STDIN Function ----- -------- 3 arr[][] sizes n = 3, m = 3 11 2 4 arr = [[11, 2, 4], [4, 5, 6], [10, 8, -12]] 4 5 6 10 8 -12	15

Explanation

The primary diagonal is:

```
11
   5
     -12
```

Sum across the primary diagonal: $11 + 5 - 12 = 4$.

The secondary diagonal is:

```
     4
   5
10
```

Sum across the secondary diagonal: $4 + 5 + 10 = 19$
Difference: $|4 - 19| = 15$

Note: |x| is the absolute value of x.

Problem G. Plus Minus

OS Linux

Given an array of integers, calculate the ratios of its elements that are $positive$, $negative$, and $zero$. Print the decimal value of each fraction on a new line with 6 places after the decimal.

Note: This challenge introduces precision problems. The test cases are scaled to six decimal places, though answers with absolute error of up to 10^{-4} are acceptable.

Example
$arr = [1, 1, 0, -1, -1]$

There are $n = 5$ elements: two positive, two negative and one zero. Their ratios are $\frac{2}{5} = 0.400000$, $\frac{2}{5} = 0.400000$ and $\frac{1}{5} = 0.200000$. Results are printed as:

```
0.400000
0.400000
0.200000
```

Function Description

Complete the $plusMinus$ function with the following parameter(s):

- $int\ arr[n]$: an array of integers

Print
Print the ratios of positive, negative and zero values in the array. Each value should be printed on a separate line with 6 digits after the decimal. The function should not return a value.

Input Format

The first line contains an integer, n, the size of the array.
The second line contains n space-separated integers that describe $arr[n]$.

Constraints

$0 < n \leq 100$
$-100 \leq arr[i] \leq 100$

Input	Output
STDIN ----- 6 -4 3 -9 0 4 1 Function -------- arr[] size n = 6 arr = [-4, 3, -9, 0, 4, 1]	0.500000 0.333333 0.166667

Explanation

There are 3 positive numbers, 2 negative numbers, and 1 zero in the array.
The proportions of occurrence are positive: $\frac{3}{6} = 0.500000$, negative: $\frac{2}{6} = 0.333333$ and zeros: $\frac{1}{6} = 0.166667$.

Problem H. Staircase

OS Linux

Staircase detail

This is a staircase of size $n = 4$:

```
   #
  ##
 ###
####
```

Its base and height are both equal to n. It is drawn using `#` symbols and spaces. **The last line is not preceded by any spaces.**

Write a program that prints a staircase of size n.

Function Description

Complete the $staircase$ function with the following parameter(s):

- $int\ n$: an integer

Print

Print a staircase as described above. No value should be returned.
Note: The last line is not preceded by spaces. All lines are right-aligned.

Input Format

A single integer, n, denoting the size of the staircase.

Constraints

$0 < n \leq 100$.

Input	Output
6	# ## ### #### ##### ######

Explanation

The staircase is right-aligned, composed of `#` symbols and spaces, and has a height and width of $n = 6$.

Problem I. Mini-Max Sum

OS Linux

Given five positive integers, find the minimum and maximum values that can be calculated by summing exactly four of the five integers. Then print the respective minimum and maximum values as a single line of two space-separated long integers.

Example
$arr = [1, 3, 5, 7, 9]$

The minimum sum is $1 + 3 + 5 + 7 = 16$ and the maximum sum is $3 + 5 + 7 + 9 = 24$. The function prints

```
16 24
```

Function Description

Complete the $miniMaxSum$ function with the following parameter(s):

- $arr[5]$: an array of 5 integers

Print

Print two space-separated integers on one line: the minimum sum and the maximum sum of 4 of 5 elements.No value should be returned.

Note For some languages, like C, C++, and Java, the sums may require that you use a long integer due to their size.

Input Format

A single line of five space-separated integers.

Constraints

$1 \leq arr[i] \leq 10^9$

Input	Output
1 2 3 4 5	10 14

Explanation

The numbers are 1, 2, 3, 4, and 5. Calculate the following sums using four of the five integers:

1. Sum everything except 1, the sum is $2 + 3 + 4 + 5 = 14$.
2. Sum everything except 2, the sum is $1 + 3 + 4 + 5 = 13$.
3. Sum everything except 3, the sum is $1 + 2 + 4 + 5 = 12$.
4. Sum everything except 4, the sum is $1 + 2 + 3 + 5 = 11$.
5. Sum everything except 5, the sum is $1 + 2 + 3 + 4 = 10$.

Hints: Beware of integer overflow! Use a 64-bit integer to store the sums.

Need help to get started? Try the Solve Me First problem.

Problem J. Birthday Cake Candles

OS Linux

You are in charge of the cake for a child's birthday. It will have one candle for each year of their total age. They will only be able to blow out the tallest of the candles. Your task is to count how many candles are the tallest.

Example

$candles = [4, 4, 1, 3]$

The tallest candles are `4` units high. There are `2` candles with this height, so the function should return `2`.

Function Description

Complete the function $birthdayCakeCandles$ with the following parameter(s):

- $int\ candles[n]$: the candle heights

Returns

- int: the number of candles that are tallest

Input Format

The first line contains a single integer, n, the size of $candles[]$.
The second line contains n space-separated integers, where each integer i describes the height of $candles[i]$.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq candles[i] \leq 10^7$

Input	Output
4 3 2 1 3	2

Explanation 0

Candle heights are $[3, 2, 1, 3]$. The tallest candles are 3 units, and there are 2 of them.

Problem K. Time Conversion

OS Linux

Given a time in 12-hour AM/PM format, convert it to military (24-hour) time.

Note: - 12:00:00AM on a 12-hour clock is 00:00:00 on a 24-hour clock.
- 12:00:00PM on a 12-hour clock is 12:00:00 on a 24-hour clock.

Example

- $s = \text{'12:01:00PM'}$

  Return '12:01:00'.

- $s = \text{'12:01:00AM'}$

  Return '00:01:00'.

Function Description

Complete the *timeConversion* function with the following parameter(s):

- *string* s: a time in 12 hour format

Returns

- *string*: the time in 24 hour format

Input Format

A single string s that represents a time in 12-hour clock format (i.e.: hh:mm:ssAM or hh:mm:ssPM).

Constraints

- All input times are valid

Input	Output
07:05:45PM	19:05:45

Problem L. Number Line Jumps

OS Linux

You are choreographing a circus show with various animals. For one act, you are given two kangaroos on a number line ready to jump in the positive direction (i.e, toward positive infinity).

- The first kangaroo starts at location $x1$ and moves at a rate of $v1$ meters per jump.
- The second kangaroo starts at location $x2$ and moves at a rate of $v2$ meters per jump.

You have to figure out a way to get both kangaroos at the same location at the same time as part of the show. If it is possible, return `YES`, otherwise return `NO`.

Example
$x1 = 2$
$v1 = 1$
$x2 = 1$
$v2 = 2$

After one jump, they are both at $x = 3$, ($x1 + v1 = 2 + 1$, $x2 + v2 = 1 + 2$), so the answer is `YES`.

Function Description

Complete the function *kangaroo* in the editor below.

kangaroo has the following parameter(s):

- *int x1, int v1*: starting position and jump distance for kangaroo 1
- *int x2, int v2*: starting position and jump distance for kangaroo 2

Returns

- *string:* either `YES` or `NO`

Input Format

A single line of four space-separated integers denoting the respective values of $x1$, $v1$, $x2$, and $v2$.

Constraints

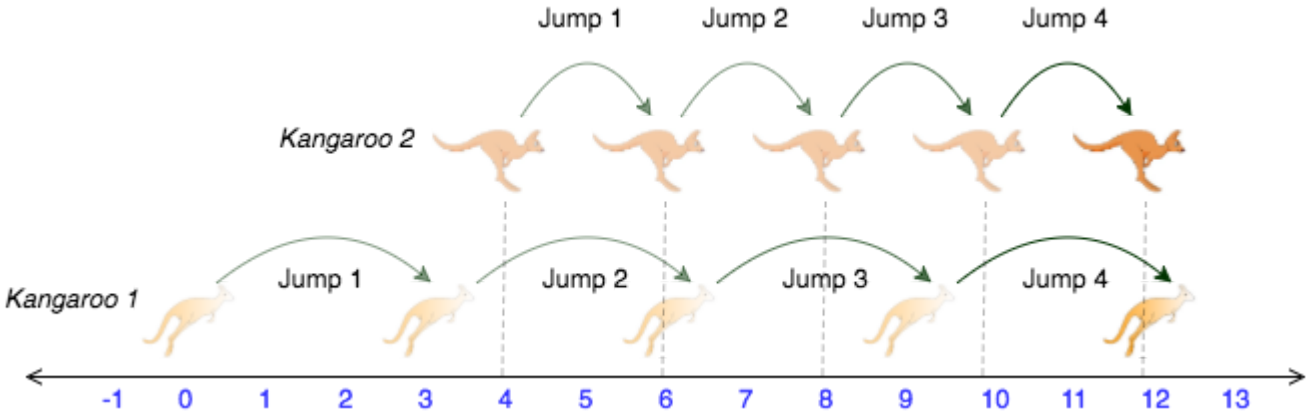
- $0 \leq x1 < x2 \leq 10000$
- $1 \leq v1 \leq 10000$

- $1 \leq v2 \leq 10000$

Input	Output
0 3 4 2	YES

Explanation 0

The two kangaroos jump through the following sequence of locations:



From the image, it is clear that the kangaroos meet at the same location (number 12 on the number line) after same number of jumps (4 jumps), and we print `YES`.

Input	Output
0 2 5 3	NO

Explanation 1

The second kangaroo has a starting location that is ahead (further to the right) of the first kangaroo's starting location (i.e., $x_2 > x_1$). Because the second kangaroo moves at a faster rate (meaning $v_2 > v_1$) *and* is already ahead of the first kangaroo, the first kangaroo will never be able to catch up. Thus, we print *NO*.

Labayk Allahuma Labayk. Labayk La shareeka laka Labayk. Innal hamda wannimata laka wal mulk. La shareeka Lak

(Here I am at your service, oh Lord, here I am - here I am. No partner do you have. Here I am. Truly, the praise and the favor are yours, and the dominion. No partner do you have.)



These are the words chanted by some two million people from world heading, as if pulled by a magnet, to one single spot on Earth. As has happened every year for 14 centuries, Muslim pilgrims gather in Makkah to perform rituals based on those conducted by the Prophet Muhammad during his last visit to the city.

Performing these rituals, known as the Hajj, is the fifth pillar of Islam and the most significant manifestation of Islamic faith and unity. Undertaking the Hajj at least once is a duty for Muslims who are physically and financially able to make the journey to Makkah. The emphasis on financial ability is meant to ensure that a Muslim takes care of his family first. The requirement that a Muslim be healthy and physically capable of undertaking the pilgrimage is intended to exempt those who cannot endure the rigors of extended travel.

The pilgrimage is the religious high point of a Muslim's life and an event that every Muslim dreams of undertaking. Umrah, the lesser pilgrimage, can be undertaken at any time of the year; Hajj, however, is performed during a five-day period from the ninth through the thirteenth of Dhu Al-Hijjah, the twelfth month of the Muslim lunar calendar.

It is generally presumed that the Hajj performed on Friday is called 'Hajj-e-Akbar' and it is a superior kind of Hajj as compared with the Hajj performed on other days of the week.

But, the correct meaning of the term, as explained by a large number of the commentators of the Holy Quran is that the Umrah, which can be performed at any time throughout the year, was generally called 'Hajj-e-Asghar' (the minor Hajj). In order to distinguish hajj from Umrah the former was named 'Hajj-e-Akbar' (the greater hajj). Therefore, each and every hajj is Hajj-e-Akbar, no matter whether it is performed on Friday or on any other day. The word 'Akbar' (greater) is used only to distinguish it from Umrah which is a minor Hajj.

Input

There will be several lines in the input terminated with a line containing a single '`*`'. This last line should not be processed. Each of the lines will contain either Hajj or Umrah.

Output

For each line of the input, output either '`Hajj-e-Akbar`' or '`Hajj-e-Asghar`' in separate lines without quotations. For exact format refer to the sample.

Sample Input

```
Hajj
Umrah
Hajj
Umrah
*
```

Sample Output

```
Case 1: Hajj-e-Akbar
Case 2: Hajj-e-Asghar
Case 3: Hajj-e-Akbar
Case 4: Hajj-e-Asghar
```

Problem N. Exams

Time Limit	1000 ms
Code Length Limit	50000 B
OS	Linux

In Chefland, there are X schools, and each school has Y students.
The year end results are in and a total of Z students passed the exams.

Assuming that all students appeared for the exams, find whether the number of students who passed in Chefland was **strictly greater** than 50%.

Input Format

- The first line of input will contain a single integer T, denoting the number of test cases.
- Each test case consists of three space-separated integers X, Y, and Z, as mentioned in the statement.

Output Format

For each test case, output on a new line, `YES`, if the total number of students who passed in Chefland was strictly greater than 50%, otherwise print `NO`.

You may print each character of the string in uppercase or lowercase (for example, the strings `YES`, `yEs`, `yes`, and `yeS` will all be treated as identical).

Constraints

- $1 \leq T \leq 2 \cdot 10^4$
- $1 \leq X \leq 5$
- $1 \leq Y \leq 50$
- $0 \leq Z \leq X \cdot Y$

Sample 1

Input	Output
4 2 10 12 2 10 3 1 5 3 3 6 9	YES NO YES NO

Test case 1: The total number of students appeared were $2 \cdot 10 = 20$. The number of students passed were 12. Thus, number of students who passed are 60%, which is strictly greater than 50%.

Test case 2**:** The total number of students appeared were $2 \cdot 10 = 20$. The number of students passed were 3.
Thus, number of students who passed are 15%, which is less than 50%.

Test case 3**:** The total number of students appeared were $1 \cdot 5 = 5$. The number of students passed were 3.
Thus, number of students who passed are 60%, which is strictly greater than 50%.

Test case 4**:** The total number of students appeared were $3 \cdot 6 = 18$. The number of students passed were 9.
Thus, number of students who passed are 50%, which is equal to 50%.

Some operators checks about the relationship between two values and these operators are called relational operators. Given two numerical values your job is just to find out the relationship between them that is (i) First one is greater than the second (ii) First one is less than the second or (iii) First and second one is equal.

Input

First line of the input file is an integer t ($t < 15$) which denotes how many sets of inputs are there. Each of the next t lines contain two integers a and b ($|a|, |b| < 1000000001$).

Output

For each line of input produce one line of output. This line contains any one of the relational operators '>', '<' or '=', which indicates the relation that is appropriate for the given two numbers.

Sample Input

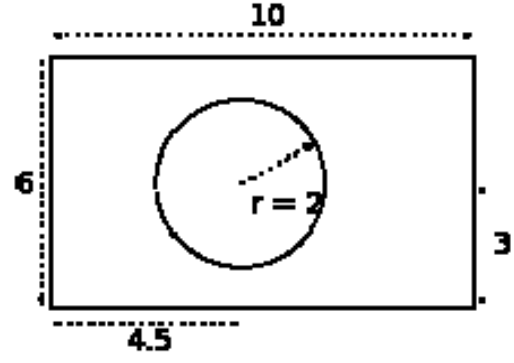
```
3
10 20
20 10
10 10
```

Sample Output

```
<
>
=
```

The national flag of Bangladesh is bottle green in color and rectangular in size with the **length** (L) to **width** ratio of 10:6. It bears a red circle on the background of green. It maintains the **length** (L) to **radius** ratio of 5:1 (If the **length** is 10 then **width** should be 6 and **radius** should be 2). The color in the background represents the greenery of Bangladesh while the red circle symbolizes the rising sun and the sacrifice of lives in our freedom fight.





Input

First line of input will contain the number of test cases, $T \leq 100$. Then there follows T lines, each containing a positive integer $L \leq 1000$, representing length of the flag.

Output

For each test case output is a line with two space separated real numbers containing exactly two digits after decimal point. Two numbers represent the area of red and green portion respectively.

Note: Pi is considered to be $\arccos(-1)$.

Sample Input

```
1
10
```

Sample Output

```
12.57 47.43
```

Problem Q. Fair Share Settlement

Time Limit	1000 ms
Code Length Limit	50000 B
OS	Linux

You and K of your friends gather for dinner at a restaurant, and the total bill is Rs. N. You pay the entire bill, and since everyone shared the food equally, the bill needs to be split evenly among you all.

After asking your friends to reimburse their fair shares, each person pays you back. However, if the amount to be paid back wasn't an integer, they round down to the previous integer value.
For example, if each friend should have paid Rs. 13.2, they instead pay Rs. 13. Even if they should have paid Rs. 13.79, they still only pay Rs. 13, rounding down.

Calculate the net amount that you ended up paying after everyone settles their debts.

Input Format

- The first line of input will contain a single integer T, denoting the number of test cases.
- Each test case consists of one line of input, containing two space-separated integers N and K.

Output Format

For each test case, output on a new line the net amount you paid.

Constraints

- $1 \leq T \leq 10^4$
- $1 \leq N \leq 10^3$
- $1 \leq K \leq 10$

Sample 1

Input	Output
3 500 1 999 9 1 4	250 108 1

Test case 1: You and another friend have gone for dinner, and the total cost has come to Rs. 500. You paid the entire amount of Rs. 500, and then your friend repaid you by paying back Rs. 250. Thus your net payment was $(500 - 250)$ = Rs. 250.

Test case 2: Each friend's fair share was 99.9. However they round down to 99 instead. Thus your net payment was $999 - 9 \cdot 99$ = Rs. 108.

Test case 3: Each friend's fair share was 0.2, so they paid you 0. So, the entire Rs. 1 is your net payment.

Write a program that will remove all duplicates from a sequence of integers and print the list of unique integers occuring in the input sequence, along with the number of occurences of each.

Input

The input file will contain a sequence of integers (positive, negative, and/or zero). The input file may be arbitrarily long.

Output

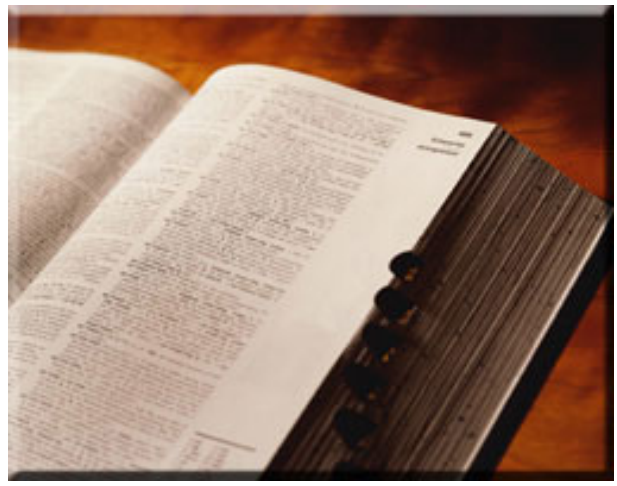
The output for this program will be a sequence of ordered pairs, separated by newlines. The first element of the pair must be an integer from the input file. The second element must be the number of times that that particular integer appeared in the input file. The elements in each pair are to be separated by space characters. The integers are to appear in the order in which they were contained in the input file.

Sample Input

```
3 1 2 2 1 3 5 3 3 2
```

Sample Output

```
3 4
1 2
2 3
5 1
```



Andy, 8, has a dream - he wants to produce his very own dictionary. This is not an easy task for him, as the number of words that he knows is, well, not quite enough. Instead of thinking up all the words himself, he has a briliant idea. From his bookshelf he would pick one of his favourite story books, from which he would copy out all the distinct words. By arranging the words in alphabetical order, he is done! Of course, it is a really time-consuming job, and this is where a computer program is helpful.

You are asked to write a program that lists all the different words in the input text. In this problem, a word is defined as a consecutive sequence of alphabets, in upper and/or lower case. Words with only one letter are also to be considered. Furthermore, your program must be CaSe InSeNsItIvE. For example, words like “Apple”, “apple” or “APPLE” must be considered the same.

Input

The input file is a text with no more than 5000 lines. An input line has at most 200 characters. Input is terminated by EOF.

Output

Your output should give a list of different words that appears in the input text, one in a line. The words should all be in lower case, sorted in alphabetical order. You can be sure that he number of distinct words in the text does not exceed 5000.

Sample Input

```
Adventures in Disneyland

Two blondes were going to Disneyland when they came to a fork in the
road. The sign read: "Disneyland Left."

So they went home.
```

Sample Output

```
a
adventures
blondes
came
disneyland
fork
going
home
in
left
read
road
sign
so
the
they
to
two
went
were
when
```

Prince Macaw is flying here and there and searching for food. Her Princess is waiting for him at home. But he stopped when he was flying near Shahbag. He saw millions of people gathered there for only one demand- death sentence of war crimes. They were giving slogans. If you are not familiar with slogans let me tell you, slogan has two parts. One part of the slogan is yelled by the leader and the rest part is chorused from the mass people. For example, when the leader says — "tomar amar thikana" then other say "padma meghna jomuna". Some of the slogans are listed below-

First Line	Second Line
ko te kader molla	tui rajakar tui rajakar
tumi ke ami ke	garo chakma bangali

Prince also joined with this mass people. When it was too late, Princess got worried about Prince. So she also went out. After some while, she found Prince on a tree at Shahbag yelling slogans in strong voice. Instead of getting mad, Princess asked Prince what are the slogans. So Prince told her a few slogans, first line and corresponding second line. So Princess got prepared. Given the list of the slogans and also the first lines Princess going to hear, print out the second line of the slogans.

Input

First line contains number of slogans, N $(0 < N \leq 20)$ Prince is going to teach Princess. Hence follows N pairs of lines. First line of the pair contains first line of a slogan and the second line of the pair is corresponding second line of the slogan. Then another positive integer Q will be given $(Q \leq 100)$. Hence follows Q lines. Each line contains first line of a slogan. You may assume that the slogans will contain lower case English alphabet and/or space. Both the lines of slogan will contain at least one alphabet. No two first lines in the Princes slogan list will be same. Each slogan will be at most 100 characters long. All of the query first line will be in Princes list.

Output

For each of the first line of the slogan, you need to print out corresponding second line.

Sample Input

```
3
ko te kader molla
tui rajakar tui rajakar
tumi ke ami ke
garo chakma bangali
jalo re jalo
agun jalo
2
jalo re jalo
ko te kader molla
```

Sample Output

```
agun jalo
tui rajakar tui rajakar
```

Problem U. Repetitions

Time Limit 1000 ms
Mem Limit 524288 kB

You are given a DNA sequence: a string consisting of characters A, C, G, and T. Your task is to find the longest repetition in the sequence. This is a maximum-length substring containing only one type of character.

Input

The only input line contains a string of n characters.

Output

Print one integer: the length of the longest repetition.

Constraints

- $1 \le n \le 10^6$

Example

Input	Output
ATTCGGGA	3

Problem V. Weird Algorithm

Time Limit 1000 ms
Mem Limit 524288 kB

Consider an algorithm that takes as input a positive integer n. If n is even, the algorithm divides it by two, and if n is odd, the algorithm multiplies it by three and adds one. The algorithm repeats this, until n is one. For example, the sequence for $n = 3$ is as follows:

$$3 \to 10 \to 5 \to 16 \to 8 \to 4 \to 2 \to 1$$

Your task is to simulate the execution of the algorithm for a given value of n.

Input

The only input line contains an integer n.

Output

Print a line that contains all values of n during the algorithm.

Constraints

- $1 \leq n \leq 10^6$

Example

Input	Output
3	3 10 5 16 8 4 2 1

Problem W. Missing Number

Time Limit 1000 ms
Mem Limit 524288 kB

You are given all numbers between $1, 2, \ldots, n$ except one. Your task is to find the missing number.

Input

The first input line contains an integer n.

The second line contains $n - 1$ numbers. Each number is distinct and between 1 and n (inclusive).

Output

Print the missing number.

Constraints

- $2 \leq n \leq 2 \cdot 10^5$

Example

Input	Output
5 2 3 1 5	4

Problem X. Maps-STL

OS Linux

Maps are a part of the C++ STL.Maps are associative containers that store elements formed by a combination of a key value and a mapped value, following a specific order.The mainly used member functions of maps are:

- *Map Template:*

```
std::map <key_type, data_type>
```

- *Declaration:*

```
map<string,int>m; //Creates a map m where key_type is of type str
```

- *Size:*

```
int length=m.size(); //Gives the size of the map.
```

- *Insert:*

```
m.insert(make_pair("hello",9)); //Here the pair is inserted into
```

- *Erasing an element:*

```
m.erase(val); //Erases the pair from the map where the key_type i
```

- *Finding an element:*

```
map<string,int>::iterator itr=m.find(val); //Gives the iterator t
Ex: map<string,int>::iterator itr=m.find("Maps"); //If Maps is nc
```

- *Accessing the value stored in the key:*

```
To get the value stored of the key "MAPS" we can do m["MAPS"] or
```

To know more about maps click Here.

You are appointed as the assistant to a teacher in a school and she is correcting the answer sheets of the students.Each student can have multiple answer sheets.So the teacher has Q queries:

1 X Y :Add the marks Y to the student whose name is X.

2 X: Erase the marks of the students whose name is X.

3 X: Print the marks of the students whose name is X. (If X didn't get any marks print 0.)

Input Format

The first line of the input contains Q where Q is the number of queries. The next Q lines contain 1 query each.The first integer, $type$ of each query is the type of the query.If query is of type 1, it consists of one string and an integer X and Y where X is the name of the student and Y is the marks of the student.If query is of type 2 or 3,it consists of a single string X where X is the name of the student.

Constraints

$1 \leq Q \leq 10^5$

$1 \leq type \leq 3$

$1 \leq |X| \leq 6$

$1 \leq Y \leq 10^3$

Output Format

For queries of type 3 print the marks of the given student.

Input	Output
7 1 Jesse 20 1 Jess 12 1 Jess 18 3 Jess 3 Jesse 2 Jess 3 Jess	30 20 0

Problem Y. Sets-STL

OS Linux

Sets are a part of the C++ STL. Sets are containers that store unique elements following a specific order. Here are some of the frequently used member functions of sets:

- *Declaration:*

```
set<int>s; //Creates a set of integers.
```

- *Size:*

```
int length=s.size(); //Gives the size of the set.
```

- *Insert:*

```
s.insert(x); //Inserts an integer x into the set s.
```

- *Erasing an element:*

```
s.erase(val); //Erases an integer val from the set s.
```

- *Finding an element:*

```
set<int>::iterator itr=s.find(val); //Gives the iterator to the e
Ex: set<int>::iterator itr=s.find(100); //If 100 is not present t
```

To know more about sets click Here. Coming to the problem, you will be given Q queries. Each query is of one of the following three types:

$1\ x$: Add an element x to the set.
$2\ x$: Delete an element x from the set. (If the number x is not present in the set, then do nothing).
$3\ x$: If the number x is present in the set, then print "Yes"(without quotes) else print "No"(without quotes).

Input Format

The first line of the input contains Q where Q is the number of queries. The next Q lines contain 1 query each. Each query consists of two integers y and x where y is the type of the query and x is an integer.

Constraints

$1 <= Q <= 10^5$
$1 <= y <= 3$
$1 <= x <= 10^9$

Output Format

For queries of type 3 print "Yes"(without quotes) if the number x is present in the set and if the number is not present, then print "No"(without quotes).
Each query of type 3 should be printed in a new line.

Input	Output
8 1 9 1 6 1 10 1 4 3 6 3 14 2 6 3 6	Yes No No

Problem Z. Fibonacci Numbers

Time Limit 1000 ms
Mem Limit 524288 kB

Given an integer N, print the N-th Fibonacci number.

A Fibonacci series is a series of numbers in which each number is the sum of the two numbers preceding it. For this problem, you can assume that the first two numbers in the series are 1 and 1.

Below you can see the first few numbers of the series:

```
1, 1, 2, 3, 5, 8, ...
```

Input

The input will contain an integer N ($0 < N < 50$).

Output

Print the N-th Fibonacci number.

Example

Input	Output
3	2

Problem AA. Small Factorial

Time Limit	1000 ms
Code Length Limit	50000 B
OS	Linux

Write a program to find the factorial value of any number entered by the user.

Input Format

The first line contains an integer **T**, the total number of testcases. Then **T** lines follow, each line contains an integer **N**.

Output Format

For each test case, display the factorial of the given number **N** in a new line.

Constraints

- $1 \leq T \leq 1000$
- $0 \leq N \leq 20$

Sample 1

Input	Output
3 3 4 5	6 24 120

Problem AB. Make N Zero

Time Limit	1000 ms
Code Length Limit	50000 B
OS	Linux

You are given a positive integer N. In one operation you can:

- Either subtract 3 or subtract 4 from N.

Can you make N zero by performing any number of operations?

Input Format

- The first line contains a single integer T, denoting the number of test cases.
- The only line of each test case contains a positive integer N, denoting the integer which you need to make zero.

Output Format

For each test case, output `YES` if you can make N zero using any number of operations. Otherwise, print `NO`.

You may print each character of the string in uppercase or lowercase (for example, the strings `YES`, `yEs`, `yes`, and `yeS` will all be treated as identical).

Constraints

- $1 \leq T \leq 10^4$
- $1 \leq N \leq 10^4$

Sample 1

Input	Output
3 1 3 7	NO YES YES

Test case 1: You cannot make $N = 1$ zero by performing any number of operations.

Test case 2: Subtract 3 from $N = 3$ to make it zero in one operation.

Test case 3: Subtract 4 from $N = 7$ to make it $N = 3$ in one operation. Then, subtract 3 from $N = 3$ to make it zero in second operation.

In 1949 the Indian mathematician D.R. Kaprekar discovered a class of numbers called self-numbers. For any positive integer n, define $d(n)$ to be n plus the sum of the digits of n. (The d stands for *digitadition*, a term coined by Kaprekar.) For example, $d(75) = 75 + 7 + 5 = 87$. Given any positive integer n as a starting point, you can construct the infinite increasing sequence of integers n, $d(n)$, $d(d(n))$, $d(d(d(n)))$, ... For example, if you start with 33, the next number is $33 + 3 + 3 = 39$, the next is $39 + 3 + 9 = 51$, the next is $51 + 5 + 1 = 57$, and so you generate the sequence

33, 39, 51, 57, 69, 84, 96, 111, 114, 120, 123, 129, 141, ...

The number n is called a ***generator*** of $d(n)$. In the sequence above, 33 is a generator of 39, 39 is a generator of 51, 51 is a generator of 57, and so on. Some numbers have more than one generator: for example, 101 has two generators, 91 and 100. A number with ***no*** generators is a $\boldsymbol{self-number}$. There are thirteen self-numbers less than 100: 1, 3, 5, 7, 9, 20, 31, 42, 53, 64, 75, 86, and 97.

Write a program to output all positive self-numbers less than or equal 1000000 in increasing order, one per line.

Sample Output

```
1
3
5
7
9
20
31
42
53
64
 |
 |       <-- a lot more numbers
 |
9903
9914
9925
9927
9938
9949
9960
9971
9982
9993
 |
 |
 |
```

Problem AD. Id and Ship

Time Limit	1000 ms
Code Length Limit	50000 B
OS	Linux

Write a program that takes in a letterclass ID of a ship and display the equivalent string class description of the given ID. Use the table below.

Class ID	Ship Class
B or b	BattleShip
C or c	Cruiser
D or d	Destroyer
F or f	Frigate

Input Format

The first line contains an integer **T**, the total number of testcases. Then **T** lines follow, each line contains a character.

Output Format

For each test case, display the Ship Class depending on ID, in a new line.

Constraints

- $1 \leq T \leq 1000$

Sample 1

Input	Output
3 B c D	BattleShip Cruiser Destroyer

Problem AE. Valid Triangles

Time Limit	1000 ms
Code Length Limit	50000 B
OS	Linux

Write a program to check whether a triangle is valid or not, when the three angles of the triangle are the inputs. A triangle is valid if the sum of all the three angles is equal to 180 degrees.

Input Format

The first line contains an integer **T**, the total number of testcases. Then **T** lines follow, each line contains three angles **A**, **B** and **C**, of the triangle separated by space.

Output Format

For each test case, display 'YES' if the triangle is valid, and 'NO', if it is not, in a new line.

Constraints

- $1 \leq \mathbf{T} \leq 1000$
- $1 \leq \mathbf{A,B,C} \leq 180$

Sample 1

Input	Output
3 40 40 100 45 45 90 180 1 1	YES YES NO

Problem AF. Opposite Task

Time Limit 1000 ms
Mem Limit 65536 kB

This problem gives you a flavor of the concept of a special judge. That means the judge is smart enough to verify your code even though it may print different results. In this problem, you are asked to find the opposite task of the previous problem.

To be specific, I have two computers where I stored my problems. Now I know the total number of problems is **n**. There are no duplicate problems, and there can be at most **10** problems in each computer. You have to find the number of problems on each of the computers.

Since there can be multiple solutions. Any valid solution will do.

Input

Input starts with an integer **T (≤ 25)**, denoting the number of test cases.

Each case starts with a line containing an integer **n (0 ≤ n ≤ 20)** denoting the total number of problems.

Output

For each case, print the number of problems stored in each computer in a single line. A single space should separate the non-negative integers.

Sample

Input	Output
3 10 7 7	0 10 0 7 1 6

Problem AG. Lift

Time Limit 1000 ms
Mem Limit 65536 kB

All of you must have noticed that the lift of AIUB is not available for students. But since you deny obeying usual rules, you always use this lift no matter what happens!

Now one day you were sleeping in the class and when you woke up you found none in the department except the guard who was in a deep sleep in his room. But luckily you found the lift on. So, you want to go to the ground floor using the lift. But the lift can be in a different floor. Then you must wait for the lift to come to your floor. Assume that it takes 4 seconds for the lift to go from any floor to its adjacent floor (up or down). It takes 3 seconds to open or close the door and you need 5 seconds to enter or exit the lift. Given your position and the lift's position you have to calculate the time for you to reach the ground floor (floor 0). Reaching ground floor means you must get out of the lift in ground floor.

Input

Input starts with an integer **T (≤ 25)**, denoting the number of test cases.

Each case contains two integers. The first integer means your position (different than 0) and the second integer means the position of the lift. Assume that the department has **100** floors (may be one day it will be :D).

Output

For each case, print the case number and the calculated time in seconds.

Sample

Input	Output
3 1 2 3 10 5 5	Case 1: 27 Case 2: 59 Case 3: 39

A student from ITESM Campus Monterrey plays with a new encryption method for numbers. These method consist of the following steps:

Steps : Example

1. Read the number N to encrypt : $M = 265$
2. Interpret N as a decimal number : $X_1 = 265$ (decimal)
3. Convert the decimal interpretation of N to its binary representation : $X_1 = 100001001$ (binary)
4. Let b_1 be equal to the number of 1's in this binary representation : $b_1 = 3$
5. Interpret N as a Hexadecimal number : $X_2 = 265$ (hexadecimal)
6. Convert the hexadecimal interpretation of N to its binary representation : $X_2 = 1001100101$
7. Let b_2 be equal to the number of 1's in the last binary representation : $b_2 = 5$
8. The encryption is the result of M *xor* $(b1 * b2)$: 265 *xor* (3*5) = 262

This student failed Computational Organization, thats why this student asked the judges of ITESM Campus Monterrey internal ACM programming Contest to ask for the numbers of 1's bits of this two representations so that he can continue playing.

You have to write a program that read a *Number* and give as output the number b_1 and b_2

Input

The first line will contain a number N which is the number of cases that you have to process. Each of the following N Lines ($0 < N \leq 1000$) will contain the number M ($0 < M \leq 9999$, in decimal representation) which is the number the student wants to encrypt.

Output

You will have to output N lines, each containing the number b_1 and b_2 in that order, separated by one space corresponding to that lines number to crypt

Sample Input

```
3
265
111
1234
```

Sample Output

```
3 5
6 3
5 5
```

Problem AI. Enormous Input Test

Time Limit	8000 ms
Code Length Limit	50000 B
OS	Linux

You are given N integers. Find the count of numbers divisible by K.

Input Format

The input begins with two positive integers N, K. The next N lines contains one positive integer each denoted by A_i.

Output Format

Output a single number denoting how many integers are divisible by K.

Constraints

- $1 \leq N, K \leq 10^7$
- $1 \leq A_i \leq 10^9$

Sample 1

Input	Output
7 3 1 51 966369 7 9 999996 11	4

The integers divisible by 3 are $51, 966369, 9$, and 999996. Thus, there are 4 integers in total.

Problem AJ. Chef And Coloring

Time Limit	1000 ms
Code Length Limit	50000 B
OS	Linux

Read problems statements in Mandarin Chinese, Russian and Vietnamese as well.

After a long time, Chef has finally decided to renovate his house. Chef's house has **N** rooms in it numbered from 1 to **N**. Each room is currently painted in one of the red, blue or green colors. Your are given configuration of colors of his house by a string **S** consisting of **N** characters. In this string, color red will be denoted by 'R', green by 'G' and blue by 'B'.

Chef does not like current painting configuration that much and would like to repaint the house such that each room has same color.

For painting, Chef has all the 3 color paints available and mixing any 2 color paints will result into 3rd color paint i.e

- R + B = G
- B + G = R
- G + R = B

For example, painting a room having red color before with green color paint will make the color of room blue.

Also, Chef has many buckets of paint of each color. Simply put, you can assume that he will not run out of paint.

Being extraordinary lazy, our little chef does not want to work much and therefore, he has asked you to find the minimum number of rooms he has to repaint **(possibly zero)** in order to have all the rooms with same color. Can you please help him?

Input

First line of input contains a single integer **T** denoting the number of test cases. First line of each test case contains an integer **N** denoting the number of rooms in the chef's house. Next line of each test case contains a string **S** denoting the current color configuration of rooms.

Output

For each test case, Print the minimum number of rooms need to be painted in order to have all the rooms painted with same color i.e either red, blue or green.

Constraints

$1 \leq T \leq 10$

$1 \leq N \leq 10^5$

$S_i = \{'R','G','B'\}$

Scoring

- Subtask 1 (40 points) : **$1 \leq N \leq 10$**
- Subtask 2 (60 points) : **original constraints**

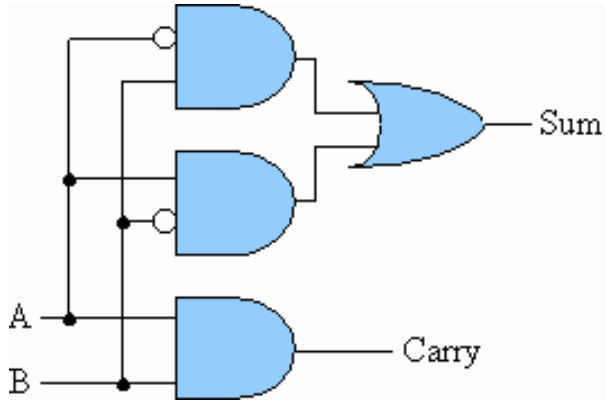
Sample 1

Input	Output
3 3 RGR 3 RRR 3 RGB	1 0 2

- **Test 1:** Chef prefers to paint room 2 with blue color such that the resulting color will be red and all the rooms have same color i.e red.
- **Test 2:** Given configuration has all the rooms painted with red color and therefore, chef does not need to do painting work at all.
- **Test 3:** One possible way of renovation is to paint room 1 with green color, room 2 with red color such that all rooms have same color i.e blue.

6+9=15 seems okay. But how come 4+6=2?

You see, Mofiz had worked hard throughout his digital logic course, but when he was asked to implement a 32 bit adder for the laboratory exam, he did some mistake in the design part. After tracing the design for half an hour, he found his flaw!! He was doing bitwise addition but his carry bit always had zero output. Thus,




```
 4 = 00000000 00000000 00000000 00000100
                     +6
   = 00000000 00000000 00000000 00000110
                       -------------------------------------
                     2 = 00000000 00000000 00000000 00000010
```

Its a good thing that he finally found his mistake, but it was too late. Considering his effort throughout the course, the instructor gave him one more chance. Mofiz has to write an efficient program that would take *2 unsigned 32 bit decimal numbers* as input, and produce an *unsigned 32 bit decimal number* as the output adding in the same was as his circuit does.

Input

In each line of input there will be a pair of integer separated by a single space. Input ends at EOF.

Output

For each line of input, output one line — the value after adding the two numbers in the '*Mofiz way*'.

Sample Input

```
4 6
6 9
```

Sample Output

```
2
15
```

Problem AL. Total Expenses

Time Limit	1000 ms
Code Length Limit	50000 B
OS	Linux

While purchasing certain items, a discount of 10% is offered if the quantity purchased is more than 1000.
If the quantity and price per item are input, write a program to calculate the total expenses.

Input Format

The first line contains an integer **T**, total number of test cases. Then follow **T** lines, each line contains integers **quantity** and **price**.

Output Format

For each test case, output the total expenses while purchasing items, in a new line.

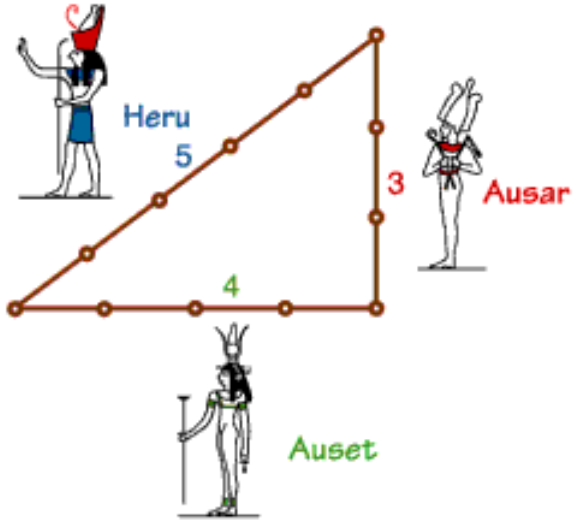
Constraints

- $1 \le T \le 1000$
- $1 \le quantity, price \le 100000$

Sample 1

Input	Output
3 100 120 10 20 1200 20	12000.000000 200.000000 21600.000000

A long time ago, the Egyptians figured out that a triangle with sides of length 3, 4, and 5 had a right angle as its largest angle. You must determine if other triangles have a similar property.



Input

Input represents several test cases, followed by a line containing '`0 0 0`'. Each test case has three positive integers, less than 30,000, denoting the lengths of the sides of a triangle.

Output

For each test case, a line containing '`right`' if the triangle is a right triangle, and a line containing '`wrong`' if the triangle is not a right triangle.

Sample Input

```
6 8 10
25 52 60
5 12 13
0 0 0
```

Sample Output

```
right
wrong
right
```

Problem AN. Probability

Time Limit	1000 ms
Code Length Limit	50000 B
OS	Linux

Chef has recently started attending probability class. For the first three lectures he had no homework but for the fourth lecture he had one to do. The problem was given as range **[A,B]** over positive integers, chef has to find the probability of a number chosen at random within the range (including **A** and **B**) such that it will contain at least one zero.

Input

- The first line of the input contains an integer **T** denoting the number of test cases. The description of **T** test cases follows.
- Each line of the test case contains two space separated integers **A** and **B** denoting the range extremities.

Output

- For each test case, output a single line containing the probability in fraction and **NOT** in decimal.

Constraints

- $1 \leq T \leq 10$
- $0 \leq A \leq 10^5$
- $0 \leq B \leq 10^6$

Example

Input:

```
2
1 20
100 999
```

Output:

```
2/20
```

```
171/900
```

Explanation

Example case 1. The range given is 1 to 20. So look at all the integers from 1 to 20, inclusive, we find that 10 and 20 are the only two numbers that have a zero in them. And hence the probability to find them is 2/20.

In a galaxy far far away there is an ancient game played among the planets. The specialty of the game is that there is no limitation on the number of players in each team, as long as there is a captain in the team. (The game is totally strategic, so sometimes less player increases the chance to win). So the coaches who have a total of N players to play, selects K $(1 \leq K \leq N)$ players and make one of them as the captain for each phase of the game. Your task is simple, just find in how many ways a coach can select a team from his N players. Remember that, teams with same players but having different captain are considered as different team.

Input

The first line of input contains the number of test cases $T \leq 500$. Then each of the next T lines contains the value of N $(1 \leq N \leq 10^9)$, the number of players the coach has.

Output

For each line of input output the case number, then the number of ways teams can be selected. You should output the result *modulo* 1000000007.

For exact formatting, see the sample input and output.

Sample Input

```
3
1
2
3
```

Sample Output

```
Case #1: 1
Case #2: 4
Case #3: 12
```

Given a list of rectangles and a list of points in the x-y plane, determine for each point which figures (if any) contain the point.

Input

There will be $n(\leq 10)$ rectangles descriptions, one per line. The first character will designate the type of figure ("`r`" for rectangle). This character will be followed by four real values designating the x-y coordinates of the upper left and lower right corners.

The end of the list will be signalled by a line containing an asterisk in column one.

The remaining lines will contain the x-y coordinates, one per line, of the points to be tested. The end of this list will be indicated by a point with coordinates 9999.9 9999.9; these values should not be included in the output.

Points coinciding with a figure border are not considered inside.

Output

For each point to be tested, write a message of the form:

`Point` i `is contained in figure` j

for each figure that contains that point. If the point is not contained in any figure, write a message of the form:

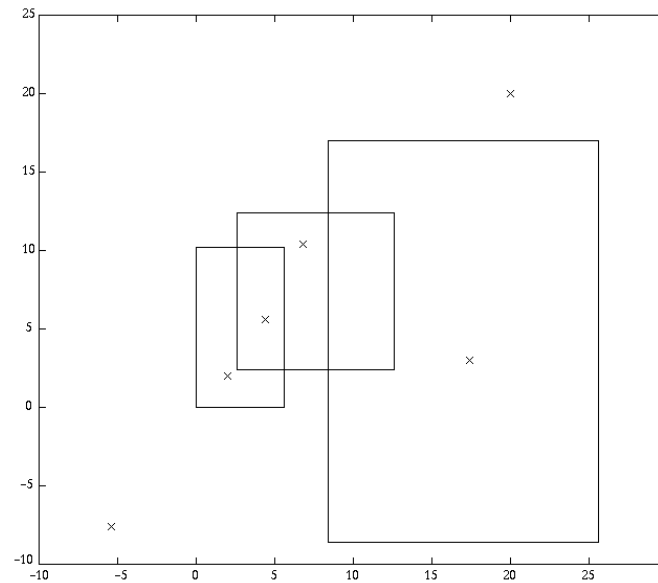
`Point` i `is not contained in any figure`

Points and figures should be numbered in the order in which they appear in the input.

Note: See the picture on the right for a diagram of these figures and data points.

Sample Input

```
r 8.5 17.0 25.5 -8.5
r 0.0 10.3 5.5 0.0
r 2.5 12.5 12.5 2.5
*
2.0 2.0
4.7 5.3
6.9 11.2
20.0 20.0
17.6 3.2
-5.2 -7.8
9999.9 9999.9
```

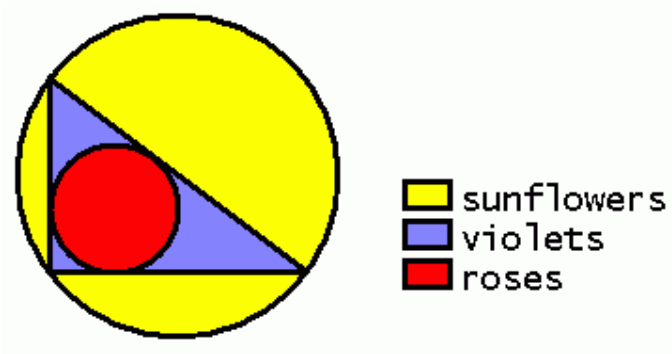


Sample Output

```
Point 1 is contained in figure 2
Point 2 is contained in figure 2
Point 2 is contained in figure 3
Point 3 is contained in figure 3
Point 4 is not contained in any figure
Point 5 is contained in figure 1
Point 6 is not contained in any figure
```

"Roses are red, violets are blue..."

Millionaire Mr Smith is well-known — not for his wealth, but for his odd sense of "art"... Mr Smith has got a circular garden. On the boundary he picks three points and gets a triangle. He then finds the largest circle in that triangular region. So he gets something like this:



Mr Smith then plants yellow sunflowers, blue violets and red roses in the way shown in the figure. (Nice combination, eh? :-) Given the lengths of the three sides of the triangle, you are to find the areas of the regions with each kind of flowers respectively.

Input

Each line of input contains three integers a, b, c, the lengths of the three sides of the triangular region, with $0 < a \leq b \leq c \leq 1000$.

Output

For each case, your program should output the areas of the regions with sunflowers, with violets and with roses respectively. Print your answers correct to 4 decimal places.



Sample Input

```
3 4 5
```

Sample Output

```
13.6350 2.8584 3.1416
```

Problem AR. Is Anagram

Time Limit 1000 ms
Mem Limit 524288 kB

An anagram is a word formed by rearranging the letters of another word.

The words "listen" and "silent" has the same letters in them appearing the same number of times. This makes these two words anagram to each other.

Given two words, you need to determine if they are anagrams of each other.

Input

The input will contain two strings A and B ($0 < \texttt{Length of A and B} < 100$), one per line.

A and B will contain lowercase alphabets only.

Output

Print `Yes` if the two words are anagrams of each other, otherwise `No`.

Example

Input	Output
listen silent	Yes

Problem AS. Fibsieve's Fantabulous Birthday

Time Limit 1000 ms
Mem Limit 65536 kB

Fibsieve had a fantabulous (yes, it's an actual word) birthday party this year. He had so many gifts that he was actually thinking of not having a party next year. Among these gifts there was an **N x N** glass chessboard that had a light in each of its cells. When the board was turned on a distinct cell would light up every second, and then go dark.

The cells would light up in the sequence shown in the diagram. Each cell is marked with the second in which it would light up.

25	24	23	22	21
10	11	12	13	20
9	8	7	14	19
2	3	6	15	18
1	4	5	16	17

In the first second the light at cell (1, 1) would be on. And in the 5th second the cell (3, 1) would be on. Now, Fibsieve is trying to predict which cell will light up at a certain time (given in seconds). Assume that **N** is large enough.

Input

Input starts with an integer **T (≤ 200)**, denoting the number of test cases.

Each case will contain an integer **S ($1 \le S \le 10^{15}$)** which stands for the time.

Output

For each case you have to print the case number and two numbers **(x, y)**, the column and the row number.

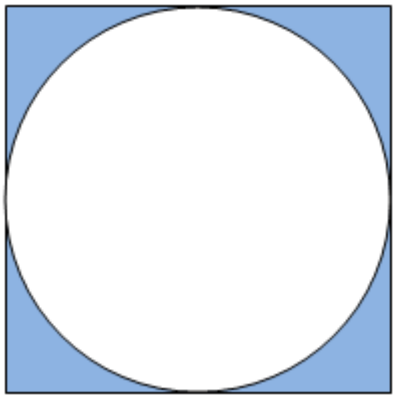
Sample

Input	Output
3 8 20 25	Case 1: 2 3 Case 2: 5 4 Case 3: 1 5

Problem AT. Circle in Square

Time Limit 1000 ms
Mem Limit 65536 kB

A circle is placed perfectly into a square. The term perfectly placed means that each side of the square is touched by the circle, but the circle doesn't have any overlapping part with the square. See the picture below.



Now you are given the radius of the circle. You have to find the area of the shaded region (blue part). Assume that **pi = 2 * acos (0.0) (acos means cos inverse)**.

Input

Input starts with an integer **T (≤ 1000)**, denoting the number of test cases.

Each case contains a floating point number **r (0 < r ≤ 1000)** denoting the radius of the circle. And you can assume that **r** contains at most four digits after the decimal point.

Output

For each case, print the case number and the shaded area rounded to two places after the decimal point.

Sample

Input	Output
3 20 30.091 87.0921	Case 1: 343.36 Case 2: 777.26 Case 3: 6511.05

Note

This problem doesn't have a special judge. So, precision problems could occur. Better to add a small value to your result to avoid this. For example, add 10^{-9} to your result.

More about rounding errors.

Problem AU. Higher Math

Time Limit 1000 ms
Mem Limit 65536 kB

You are building a house. You'd prefer if all the walls have a precise right angle relative to the ground, but you have no device to measure angles. A friend that says he has a great idea how you could ensure that all walls are upright: All you need to do is step away a few feet from the wall, measure how far away you are from the wall, measure the height of the wall, and the distance from the upper edge of the wall to where you stand. You friend tells you to do these measurements for all walls, then he'll tell you how to proceed. Sadly, just as you are done, a timber falls on your friend, and an ambulance brings him to the hospital. This is too bad, because now you have to figure out what to do with your measurements yourself.

Given the three sides of a triangle, determine if the triangle is a right triangle, i.e. if one of the triangle's angles is **90** degrees.

Input

Input starts with an integer **T (≤ 200)**, denoting the number of test cases.

Each test case consists of three integers **1 ≤ a, b, c ≤ 40000** separated by a space. The three integers are the lengths of the sides of a triangle.

Output

For each case, print the case number and `yes` or `no` depending on whether it's a right angle or not.

Sample

Input	Output
2 36 77 85 40 55 69	Case 1: yes Case 2: no

Problem AV. How Cow

Time Limit 1000 ms
Mem Limit 65536 kB

Mr Kopa Samsu is a farmer. He has a land of rectangular shape. But there are cows that disturb him a lot. The cows use to enter his land and ruin his crops. Now Mr Kopa Samsu has become smarter. He has a GPS system that will help him to know the position of the cows. So, you can think his land as a 2D grid, and cows can be treated as points. Now you are given the information of his land and cows. You have to tell him whether a cow is inside his land or not.

Input

Input starts with an integer **T (≤ 50)**, denoting the number of test cases.

The first line of each case contains four integers x_1 y_1 x_2 y_2, where (x_1, y_1) is the lower left coordinate of his land and (x_2, y_2) is the upper right coordinate of his land. You can assume that the sides of the land are axis parallel. The next line contains an integer **M (1 ≤ M ≤ 100)** denoting the number of cows. Each of the next **M** lines contains two integers each denoting **x y** - the position of a cow. You can safely assume that no cow will lie on the boundary of the rectangle. All the coordinates will lie in the range **[0, 10000]**.

Output

For each case you have to print the case number in a line first. Then for each cow, you have to print `Yes` or `No` depending whether the cow is inside the land or not.

Sample

Input	Output
1 1 2 8 10 7 0 0 5 6 1 0 7 9 3 5 10 10 1 11	Case 1: No Yes No Yes Yes No No

Problem AW. IP Checking

Time Limit 1000 ms
Mem Limit 65536 kB

An IP address is a 32 bit address formatted in the following way: **a.b.c.d**

where **a, b, c, d** are integers each ranging from **0 to 255**. Now you are given two IP addresses, first one in decimal form and second one in binary form, your task is to find if they are the same or not.

Input

Input starts with an integer **T (≤ 100)**, denoting the number of test cases.

Each case starts with two lines. First line contains an IP address in decimal form, and second line contains an IP address in binary form. In binary form, each of the four parts contains 8 digits. Assume that the given addresses are valid.

Output

For each case, print the case number and `Yes` if they are same, otherwise print `No`.

Sample

Input	Output
2 192.168.0.100 11000000.10101000.00000000.11001000 65.254.63.122 01000001.11111110.00111111.01111010	Case 1: No Case 2: Yes

Problem AX. False Ordering

Time Limit 1000 ms
Mem Limit 65536 kB

We define **b** is a Divisor of a number **a** if **a** is divisible by **b**. So, the divisors of 12 are **{1, 2, 3, 4, 6, 12}**. So, 12 has 6 divisors.

Now you have to order all the integers from 1 to 1000. **x** will come before y if:

1. Number of divisors of **x** is less than number of divisors of **y**.
2. Number of divisors of **x** is equal to number of divisors of **y** and **x > y**.

Input

Input starts with an integer **T (≤ 1005)**, denoting the number of test cases.

Each case contains an integer **n (1 ≤ n ≤ 1000)**.

Output

For each case, print the case number and the $\mathbf{n}^{\mathbf{th}}$ number after ordering.

Sample

Input	Output
5 1 2 3 4 1000	Case 1: 1 Case 2: 997 Case 3: 991 Case 4: 983 Case 5: 840